# Spatiotemporal Patterns of Crime in Chicago

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## Introduction

Effective crime prevention requires optimizing resources using existing data. By identifying spatiotemporal patterns, police departments can shift from reactive measures to proactive deployment strategies that anticipate criminal activity rather than simply responding to it.

This project analyzes real crime data from Chicago (2012–2016) to characterize crime hotspots and temporal trends. Using parametric statistical tests (due to normal distribution) and ARIMA modeling, this study evaluates the stability of high-crime areas and the feasibility of short-term predictive policing.

## Objectives

The main objective is to conduct a spatiotemporal analysis of crime incidents in Chicago (2012–2016), identifying persistent hotspots and temporal trends to inform predictive modeling.

1. Identify the major temporal trends in Chicago crime.
2. Detect the most persistent crime hotspots.
3. Evaluate spatial and temporal variations across distinct crime categories.
4. Develop and validate an ARIMA time-series model to predict short-term crime counts based on historical data.

## References

[1] Currie32. Crimes in Chicago. Kaggle Dataset. (2017).

[2] Doyouevendata. Chicago Community Areas GeoJSON. Kaggle Dataset. (2017).

## Course Information

**Course:** Computational Statistics
**Professor:** Prof. Zaussinger

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## Result 1: Spatiotemporal Analysis

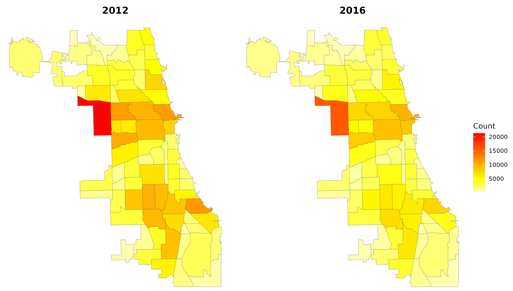



**Figure 1:** 2012 and 2016

**Temporal** analysis reveals distinct weekly patterns. While visual inspection of Figure 2 immediately highlights the huge drop in nighttime crime activity, daily variations remain slight to the naked eye. However, a One-way ANOVA confirmed significant differences in daily crime means ($F = 6.51, p < 0.001$). Post-hoc Tukey tests identified Friday as the peak day compared to the rest of the week. Additionally, a Welch Two Sample t-test confirmed a significant reduction in criminal activity between midnight and 8 AM ($t = -56.4, p < 2.2e^{-16}$), identifying this as a consistent low-risk temporal window.

As visualized in the heatmaps, the **spatial** distribution of crime incidents remained remarkably consistent across Chicago zones over the four years. A Pearson correlation analysis yielded an almost perfect coefficient ($R = 0.98, p < 0.05$), demonstrating that high-crime hotspots are persistent and static. Furthermore, when analyzing specific crime categories, a Pearson's Chi-squared test ($p < 2.2 \times 10^{-16}$) rejected the null hypothesis of independence. This confirms that different crime types do not occur randomly but are significantly clustered in distinct geographic zones.

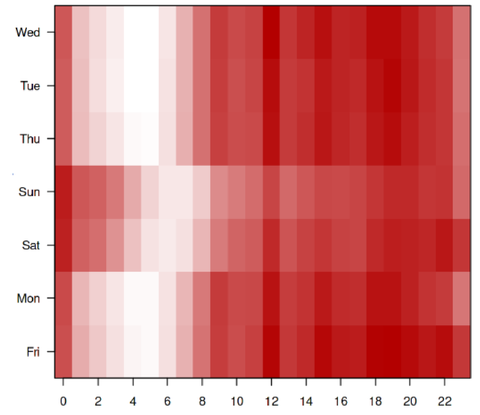



**Figure 2:** Crimes along the week

## Materials & Methods

This study analyzed a dataset of **1,419,601** reported crime incidents from the Chicago Data Portal. Data processing and visualization were performed in **R** using `tidyverse`, `lubridate`, `tidyr`, `sf`, and `forecast`.

**Statistical Validation:** The assumption of normality was validated using Q-Q plots, Shapiro-Wilk tests (p-value = 0.776), skewness (0.185), and kurtosis (3.103). Temporal patterns were assessed using One-way ANOVA and two-sample t-test. Spatial stability and category dependence were evaluated using Pearson correlation ($r$) and Pearson's Chi-squared ($\chi^2$) test. Finally, an ARIMA model was generated to forecast 2016 trends, validated using MAPE.

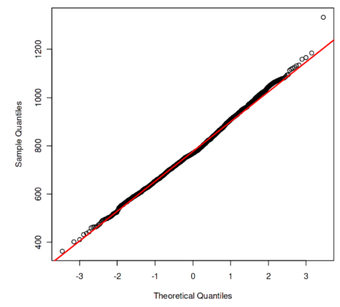
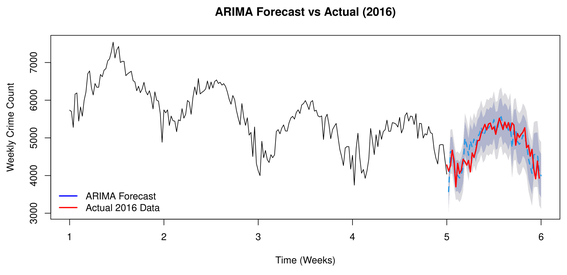



**Figure 3:** Q-Q Plot

## Result 2: Predictive Model




An ARIMA model was trained on weekly crime counts from 2012 to 2015, and used to predict the 2016 data. This model showed a high accuracy, with a **Mean Absolute Percentage Error (MAPE)** of **5.22%**, as clearly illustrated in the forecast plot.

## Conclusion

- **High Stability:** Chicago crime shows stable spatial patterns over time ($R = 0.98$), confirming that historical hotspots are reliable indicators for future resource deployment.
- **Temporal Strategy:** Significant daily and hourly differences in crime activity were identified, indicating that crime risk is not uniform, with higher activity on Fridays and during specific peak hours, and lower activity at night.
- **Targeted Intervention:** Crime types are not randomly distributed across zones ($\chi^2$ test), implying that different crime categories require geographically targeted strategies.
- **Predictive Feasibility:** The ARIMA model's accuracy (MAPE $\approx 5\%$) demonstrates that short-term crime trends can be effectively predicted using time-series analysis.

## Future Research

Future studies could improve accuracy by including factors like weather or holidays in the prediction model. Additionally, analyzing specific high-crime neighborhoods separately, rather than the whole city, would help police target their resources even more precisely.